ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

$$f_x = 3200 pixels; fieldofview = 160mm; 1pixel = 0.125mm$$

Q1(b) [3 marks]

The smallest resolvable force is 0.122 N.

Q1(c) [3 marks]

$$F = e^(AT) = e^(-0.2) = 0.818731$$

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

Q3(c) [4 marks]

Q3(d) [2 marks]

An observer uses the plant model and known control input to estimate unmeasured states such as velocity, and its poles can be designed. A simple low-pass filter cannot exploit the state-space model in this way. A filter may be safer when the model is poorly known or unreliable, because it relies less on model accuracy.

The observer uses the plant model and the measured output to reconstruct the unmeasured state. Its convergence depends on the model and the observer poles. A low-pass filter is simpler, but generally introduces phase lag and does not explicitly use the plant dynamics.

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) **[2 marks]**

The actual feedforward contribution is 0.75x8=6.

Q4(c) **[4 marks]**

$$k_1 = 18, k_2 = 6, K_I = 32$$

Q4(d) **[2 marks]**

At steady state, a stable closed loop with integral action drives $\dot{\xi}$=r-θ to zero; therefore r-θ=0 and the tracking error is zero, assuming no saturation or other implementation limits prevent convergence. A practical failure mode is integrator windup during actuator saturation; the standard remedy is anti-windup, such as back-calculation or conditional integration.

With integral action, a constant tracking error keeps accumulating in the integral state, so the controller can adjust its output until the error goes to zero, assuming the closed loop is stable and the actuator is not constrained. Under saturation the integral state can keep growing; anti-windup limits or back-calculates that state.